Exercise 1

Consider a robot manipulator with $\boldsymbol{q} \in \mathbb{R}^n$ joint variables that is redundant with respect to a task described by $\boldsymbol{r} \in \mathbb{R}^m$ variables, with $m < n$. The $m \times n$ task Jacobian matrix $\boldsymbol{J}(\boldsymbol{q})$ relates task and joint velocities, i.e., $\dot{\boldsymbol{r}} = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}}$. For task regulation problems, kinematic control typically defines a law of the form

$$\dot{\boldsymbol{q}} = \boldsymbol{J}^{\#}(\boldsymbol{q})\boldsymbol{K}\boldsymbol{e} + \left(\boldsymbol{I} - \boldsymbol{J}^{\#}(\boldsymbol{q})\boldsymbol{J}(\boldsymbol{q})\right)\dot{\boldsymbol{q}}_0, \qquad \boldsymbol{K} > 0\,(\text{diagonal}), \quad \boldsymbol{e} = \boldsymbol{r}_d - \boldsymbol{r}, \tag{1}$$

The error is $s_d - s$, where s_d are the desired positions of the points in the image plane.

The differential relation between $\dot{q}$ and $\dot{s}$ is given by the so called image Jacobian. let $e = s_d - s$.

$$\dot{s} = J(q)\dot{q} = J_p J_m \dot{q}$$

$$r = \|e\| = (e^T e)^{1/2} = (s_d^T s_d + s^T s - 2 s_d^T s)^{1/2} \Rightarrow \dot{r} = \underbrace{\frac{\partial r}{\partial q}}^{J_z} \dot{q}$$

$$\frac{\partial r}{\partial q} = -\frac{1}{2\sqrt{(s_d^T s_d + s^T s - 2 s_d^T s)}} \cdot (2 J^T s - 2 s_d^T J)$$

$$-\frac{1}{\sqrt{(s_d^T s_d + s^T s - 2 s_d^T s)}} \cdot J^T (s - s_d) = J_z \in Mat(1 \times n)$$

$$= -\frac{1}{\|e\|} J^T \underbrace{(s - s_d)}_{e}$$ The problem is that is not defined when $\|e\| = 0$. The pseudo inverse is $J_z^{\#} = -\frac{1}{\|e\|} e^T J \left(\frac{1}{\|e\|^2} J^T e e^T J\right)^2$

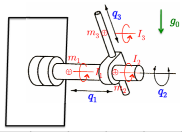
Exercise 2

For a robot of the cylindrical type with a sequence PRP of joints and mounted on a vertical wall, provide the dynamic model in the usual form

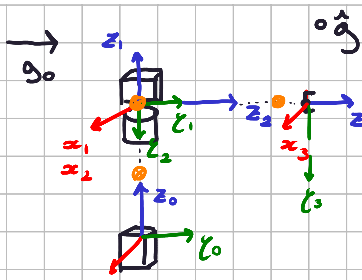
$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{F}\dot{\boldsymbol{q}} = \boldsymbol{\tau}, \tag{3}$$

using the generalized coordinates $\boldsymbol{q} \in \mathbb{R}^3$ and the dynamic coefficients defined in Fig. 1. In (3), $\boldsymbol{F} > 0$ is a diagonal matrix of viscous coefficients. Make reasonable assumptions on the zero values of the variables q_i, $i = 1, 2, 3$, and neglect the small offset between joint axes 2 and 3 (i.e., assume that these two axes intersect). Moreover, assume that the center of mass of each link is placed on the joint axis having the same index.

Without any a priori knowledge of dynamic parameters, define all the terms needed in the design of an adaptive control law for this robot so as to achieve global asymptotic tracking of a desired joint trajectory $\boldsymbol{q}_d(t)$, with $q_{di}(t) = q_{0i} + A_i(1 - \cos(2\pi t/T))^{\nu}$ $(i = 1, 2, 3)$ of amplitude A_i and period T, for an unlimited time $t \geq 0$. Which is the minimum value of the integer $\nu \in \mathbb{N}$ that allows asymptotic exact tracking?



$${}^0\hat{g} = (0 \quad g_0 \quad 0)^T$$



	α	a	d	θ
1	0	0	q_1	0
2	$-\pi/2$	0	0	q_2
3	0	0	q_2	0

$${}^1p_{c_1} = (0 \quad 0 \quad -d_1)^T$$

$${}^2p_{c_2} = 0$$

$${}^3p_{c_3} = (0 \quad 0 \quad -d_3)^T$$

$${}^0T_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2T_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1\omega_1 = 0 \qquad {}^1v_1 = (0 \quad 0 \quad \dot{q}_1)^T \Rightarrow T_1 = \frac{1}{2} m_1 \dot{q}_1^2$$

$${}^2\omega_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ 0 & 0 & -1 \\ -s_2 & c_2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{q}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -\dot{q}_2 \\ 0 \end{pmatrix} \qquad {}^2v_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ 0 & 0 & -1 \\ -s_2 & c_2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = (0 \quad -\dot{q}_1 \quad 0)^T$$

$v_{c2} = {}^2v_2 + {}^2\omega_2 \times {}^2p_{c2} = (0 \ \ -\dot q_1 \ \ 0)^T \Rightarrow T_2 = \frac{1}{2} m_2 \dot q_1^2 + \frac{1}{2} I_{2z} \dot q_2^2$

${}^3\omega_3 = (0 \ \ -\dot q_2 \ \ 0)^T \qquad {}^3v_3 = {}^2v_2 + (0 \ \ 0 \ \ \dot q_3)^T + (0 \ \ -\dot q_2 \ \ 0)^T \times (0 \ \ 0 \ \ q_3)^T$

$= (-\dot q_2 q_3 \ \ -\dot q_1 \ \ \dot q_3)^T, \quad v_{c3} = {}^3v_3 + \begin{pmatrix} 0 & 0 & -\dot q_2 \\ 0 & 0 & 0 \\ \dot q_2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -d_3 \end{pmatrix} = \begin{cases} -\dot q_2 (q_3 - d_3) \\ -\dot q_1 \\ \dot q_3 \end{cases}$

$\Rightarrow T_3 = \frac{1}{2} I_{3z} \dot q_2^2 + \frac{1}{2} m_3 \left(\dot q_2^2 (q_3 - d_3)^2 + \dot q_1^2 + \dot q_3^2 \right)$

$M_{11} = m_1 + m_2 + m_3 \qquad M_{22} = I_{2z} + I_{3z} + m_3 (d_3^2 + q_3^2 - 2 d_3 q_3) \qquad M_{33} = m_3$

$\quad = e_1 \qquad\qquad\qquad = e_2 + e_4 q_3^2 - 2 e_3 q_3 \qquad\qquad = e_4$

$M = \begin{pmatrix} e_1 & 0 & 0 \\ 0 & e_2 - 2e_3 q_3 + e_4 q_3^2 & 0 \\ 0 & 0 & e_4 \end{pmatrix}$ For the pot. energy i consider the z coordinate of the CoM's. Observ: p_{c1z}, p_{c2z} are constant.

Now i compute ${}^0p_{c3}$

${}^0T_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & q_1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

${}^0p_{c3} = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & q_1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ q_3 - d_3 \\ 1 \end{pmatrix} = \begin{cases} -s_2 (q_3 - d_3) \\ c_2 (q_3 - d_3) \\ q_1 \end{cases}$

$\Rightarrow U = m_3 g_0 c_2 (q_3 - d_3) + U_0$ const

$\frac{\partial U}{\partial q_2} = -m_3 g_0 s_2 (q_3 - d_3) \qquad \frac{\partial U}{\partial q_3} = m_3 g_0 c_2 = g_0 e_4 c_2$

$= -g_0 (m_3 s_2 q_3 - m_3 d_3 s_2) = g_0 s_2 (e_2 - e_4 q_3)$

$C_1 = 0 \Rightarrow c_1 = 0 \qquad C_2 = -e_3 + e_4 q_3 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \Rightarrow c_2 = 2 \dot q_2 \dot q_3 (e_4 q_3 - e_3)$

$C_3 = (e_3 - e_4 q_3) \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow c_3 = (e_3 - e_4 q_3) \dot q_2^2$

The model is $\qquad$ let $S_1 = e_5 \quad S_2 = e_6 \quad S_3 = e_7$

$$\begin{cases} e_1 \ddot q_1 + S_1 \dot q_1 = \tau_1 \\ (e_2 - 2 e_3 q_3 + e_4 \dot q_3^2) \ddot q_2 + 2 \dot q_2 \dot q_3 (e_4 q_3 - e_3) + g_0 s_2 (e_2 - e_4 q_3) + S_2 \dot q_2 = \tau_3 \\ e_4 \ddot q_3 + (e_3 - e_4 q_3) \dot q_2^2 + g_0 e_4 c_2 + S_3 \dot q_3 = \tau_3 \end{cases}$$

Now i search a linear parametrization:

$$Y = \begin{bmatrix} \ddot q_1 & 0 & 0 & 0 & \dot q_1 & 0 & 0 \\ 0 & \ddot q_2 + g_0 S_2 & -2q_3\ddot q_2 - 2\dot q_2 \dot q_3 & q_3^2 \ddot q_2 + 2\dot q_2 \dot q_3 q_3 - q_3 g_0 S_2 & 0 & \dot q_2 & 0 \\ 0 & 0 & q_2^2 & \ddot q_3 - q_3 \dot q_2^2 + g_0 C_2 & 0 & 0 & \dot q_3 \end{bmatrix}$$

$\tau = Ye$ in this way i consider the Following law:

let $\dot q_r = \dot q_d + \Lambda (q_d - q)$ with Λ diagonal and pos. def.

$\tau = Y\hat e + K_P e + K_D \dot e$ with K_P, K_D diagonal and pos. def.

$\dot{\hat e} = \Gamma Y^T (\dot q_r - \dot q)$ with Γ diagonal and pos. def.

Exercise 3

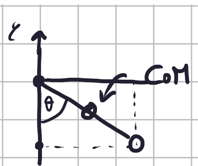
Consider an actuated pendulum that moves in the vertical plane under a joint torque u. The single link is a uniform thin rod of mass $m = 10$ kg and length $L = 2$ m. The downward equilibrium is at $\theta = 0$. Suppose that the link is initially at rest in $\theta_0 = 0$, and that we want to regulate its angular position to $\theta_d = \pi/3$ rad using the control scheme

$$u = k_p(\theta_d - \theta) - k_d\dot\theta + u_{i-1}, \qquad i = 1, 2, \ldots \tag{4}$$

with $k_p = 500$ and $k_d = 45$. The constant feedforward u_{i-1} in (4) is updated at every new reached equilibrium configuration θ_i, $i = 1, 2, \ldots$, as

$$u_i = u_{i-1} + k_p(\theta_d - \theta_i), \quad i = 1, 2, \ldots, \qquad \text{with } u_0 = 0. \tag{5}$$

Will the error $e_i = \theta_d - \theta_i$ converge to zero for $i \to \infty$ with this iterative control scheme? Why, or why not? If (or when) it does, can you predict which is the minimum number $i_{min} > 0$ of iterations guaranteeing that the error will satisfy $|e_i| < \varepsilon = 0.01$ rad for all $i \geq i_{min}$?



$P_c = \begin{cases} \sin\theta \\ -\cos\theta \end{cases}$ $\quad v_c = \begin{cases} \cos\theta\,\dot\theta \\ \sin\theta\,\dot\theta \end{cases}$ $\Rightarrow T = \frac{1}{2} I \dot\theta^2 + \frac{1}{2} m \dot\theta^2$ $\quad I = \frac{1}{3} mL^2 \Rightarrow M = \frac{70}{3} = 23.\overline{3}$

the CoM is at dist. 1 meter. The pot. energy is $U = \cos\theta\, m g_0$

$\Rightarrow g := -m g_0 \sin\theta \approx -98 \sin\theta$ the model is

$e_1 \ddot\theta - e_2 \sin\theta = u \qquad e_1 = \frac{70}{3} \qquad e_2 = 10 \cdot g_0$

To know if the system converge, we compute:

$$\frac{\partial g}{\partial \theta} = -e_2 \cos\theta \Rightarrow \max_\theta \left\|\frac{\partial g}{\partial \theta}\right\| = e_2 \approx 98 \leq 99 = \alpha$$

we set $\alpha = 99$ an upperbound for $\|\frac{\partial g}{\partial \theta}\|$.

We rewrite $K_p = 3 \cdot K \Rightarrow K = \frac{500}{3}$

The system converge to θ_d if $\lambda_{min}(K) > 99 \Rightarrow$ converge.

$\frac{500}{3}$

The reduction map is:

$$\|e_i\| < \frac{1}{2}\|e_{i-1}\|$$